

Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose

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Abstract

Deployed LLM agents are organisations of hand-offs: planner to worker to reviewer to aggregator. Every hand-off is a trust boundary, and a prompt injection that propagates to one agent can be carried forward as that agent forwards its output. Existing benchmarks measure one agent over one hop; they cannot quantify whether a compromise survives transfer, how far it travels, or where to intervene. We present *Contagion*, a measurement study of propagation in LLM agent networks over specified feed-forward graph classes (chains, star/tree-like topologies). Using two independent protocols—a *controlled per-edge* protocol that estimates conditional survival $s_i = \Pr[C_i=1 \mid C_{i-1}=1]$, and a *natural-run* protocol that measures end-to-end ASR and the empirical reproduction number R_0 —we show: (i) $ASR = \prod_i s_i^{\text{nat}}$ is an algebraic identity for chains; the key empirical question is *transportability* of isolated per-hop estimates to in-chain propagation (not guaranteed), with depth-error slope providing a diagnostic. We state independence/shared-ancestor assumptions for product-of-complements/percolation and treat such forms as exact only under those assumptions (requiring path-based adjustment or approximation when paths share ancestors). (ii) Payload *form* is a first-order mechanism: the same target survives at 0.067 arriving as a bare assertion and at 0.967 embedded in a work product. (iii) For finite acyclic (feed-forward) graphs, the standard linearised next-generation matrix is strictly triangular ($\rho(M) = 0$) as a structural property of acyclicity rather than an empirical safety guarantee; we therefore treat R_0 as a *descriptive*, *finite-horizon* reproduction statistic (with operational definition, denominator, generation convention, and finite-acyclic scope) and report $R_0 < 1$ alongside ASR of 0.45–0.73. (iv) Defence placement is structural (depends on paths/cut sets and whether a node lies on all s – t paths) rather than uniform across parallel paths; we avoid over-generalising beyond the tested topologies, and note our primary outcome is *textual propagation* (ASV) rather than automatic instruction adoption.

CCS Concepts

• **Computing methodologies** → Multiagent systems; Natural language processing; • **Security and privacy** → Social engineering.

Keywords

Multiagent systems, Prompt injection, Security measurement, Epidemic threshold, LLM agents, Benchmarks

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [18], AutoGen [35] and AgentVerse [10] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.

Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker’s intent, and that output is the *input* of the next agent, so the injection does not stay where it landed. Single-agent work establishes that one agent can be hijacked through a tool response [12, 38, 39] and that an injection can persist within one application [16]; that it also propagates agent-to-agent is now established as well [3, 24]. What has not been established is a framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in two ways that we demonstrate empirically rather than assume.

(a) *It hides where failure happens.* In a five-agent chain per-hop survival ranges from 0.133 (worker) to 0.867 (reviewer)—a 6.5× spread within one model and attack—so “the attack failed” and “the attack failed at one specific edge” are findings only the per-hop view separates.

(b) *It is not comparable across models without a no-defence baseline.* A resistant frontier model is compromised at 0.10 with *no defence at all*, so a defence driving 0.10 → 0.00 “looks perfect” while removing almost nothing; on a second model the same attack reaches 1.00 and the same defence drives it to 0.00—same defence, attack and code, opposite conclusions. We call this the *floor effect* and treat it as a first-class threat to validity.

What we build. The framework rests on three commitments: per-hop survival s_i from a *controlled per-edge* protocol (force the source compromised, feed its output to the receiver, judge the receiver) and end-to-end ASR from an *independent natural-run* protocol; an injected task semantically competitive with the agent’s legitimate assignment rather than a marker echo, which saturates at



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1.0; and reported resolution—Wilson intervals [6, 34], a bootstrap test [14] with a stated minimum detectable effect, and estimators validated on synthetic ground truth. What distinguishes this from another injection benchmark is that its central results are structural: composing per-hop rates in a chain is an *identity* plus a transportability hypothesis, and the epidemic threshold $\rho(M) < 1$ [20, 28] is *identically* satisfied on any feed-forward graph and so certifies nothing—properties of the graph and its roles, undefined for a single agent.

Findings, stated as prescriptions. We apply the framework to five models from five independent developers—coverage differs by claim (cross-model susceptibility spans all five, the depth profile three, the topology comparison three models with two instances for Llama)—and reach four conclusions, each of which changes what a practitioner should do.

- (1) **Do not compose isolated injection results into pipeline predictions without first checking that they transport.** Transportability is not guaranteed: where it fails, error *grows with distance* (+11.8 points per hop, rank correlation +1.00), and the depth-error slope is the diagnostic (Section 5.7); where it transports (within noise), $\prod s^{\text{ctrl}} \approx \text{ASR}^{\text{nat}}$. Composition of isolated estimates should state graph assumptions (e.g. parent-reach events conditionally independent, no unmodelled shared ancestors/correlations invalidating product-of-complements; product-of-complements holds under those assumptions or may require path-based adjustment when shared paths exist) and be supported by a transport check for the target graph class (Section 5.2, Section 2.1). Given our evaluation is limited to specified feed-forward graph classes (chains, star/tree-like topologies), we do not claim these prescriptions generalise to arbitrary networks without empirical validation.
- (2) **Report per-hop survival conditioned on the receiving role, together with ASR and legitimate-task retention, and always include the no-defence baseline.** Per-hop survival varies $6.5\times$ across receiver roles within one model and attack (Section 5), and a defence’s efficacy is otherwise unidentifiable (the floor effect, Section 5.5).
- (3) **Do not use $R_0 < 1$ as a safety criterion for a feed-forward agent network.** For strictly acyclic graphs the standard linearized next-generation matrix is nilpotent ($\rho(M) = 0$) as a structural property of acyclicity rather than an empirical safety guarantee; accordingly we treat R_0 as a descriptive, finite-horizon reproduction statistic (with its denominator, generation convention, scope, and limitations stated) and report $R_0 < 1$ alongside end-to-end success of 0.45–0.73 (Section 3.8/2.1). Reach and reproduction are distinct risks.
- (4) **Treat defence placement as a structural question.** Effectiveness depends on paths and cut sets (vertex/edge cuts): a hardened node is not equally effective when parallel or edge-disjoint s - t paths exist (removing one path may leave others open). Placement should be justified by whether the node lies on all paths (membership in some minimal s - t cut) for the specific graph class tested, and supported by a matched no-defence baseline, not inferred solely from per-hop survival (Section 3.8). We avoid generalising equal effectiveness

to multi-path DAGs and restrict claims to the topologies evaluated.

We also report one of our own initially significant results that proved to be an artefact of the controlled protocol, and the change that removed it (Section 5.3); in a measurement study the protocol is part of the result.

2 Related Work and Positioning

Single-agent indirect injection. Prompt injection as an attack class was first named and systematised by Pérez and Ribeiro [29], who showed that an attacker controlling any text an LLM reads can override prior instructions. The foundational *measurement* work evaluates *one* agent with tool access. InjecAgent [38] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct [36] GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [12] provides a dynamic evaluation environment for one agent over one hop, and Agent Security Bench [39] broadens this to ten application scenarios with an average attack success rate above 84%. Tool calls are prompted or learned [30, 31]; agent benchmarks [4, 19, 25] report the same scalar. Liu and Gong [26] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and introduce the ASV/MR metrics we reuse. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Liu and Gong also report that attack success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents, and the closest prior work. Greshake et al. [16] demonstrate a self-replicating injection “worm” within a single application, establishing persistence but not topology-dependent spread. Prompt Infection [24] lifts the virus metaphor to agent-to-agent spread and shows that compromised agents infect one another even under local messaging; it proposes tagging agent output to restore the user/agent provenance boundary. The nearest work is ACIArena [3], which names the threat *agent cascading injection* and benchmarks 1,356 cases over six multi-agent frameworks and three attack surfaces. MAST [7] separately catalogues multi-agent failures and attributes 32.3% of them to inter-agent misalignment, and concurrent work reports aggregate compromise rates through hierarchical chains [1].

ACIArena and Prompt Infection report *outcome* rates per attack surface or per application; we report the *conditional* probability $\Pr[C_v=1 \mid C_u=1]$ for each edge, which makes failure localisable, and test whether an isolated rate predicts its in-chain rate—a question an aggregate rate cannot pose. ACIArena also reports that narrowly scoped defences fail to transfer; we make the mechanism explicit as the *floor effect* (Section 5.5), fixed by a matched no-defence baseline.

Defences. Adaptive attacks against single-agent injection defences break eight defences at $> 50\%$ ASR, including paraphrasing, using token-level optimisation [37]; our negative result on semantic paraphrasing inside a chain is consistent with that finding. Structured queries [9], encoding [17], and removing the agent’s authority to act on untrusted text [11] instead constrain how such content

reaches the model. Our attacker instead uses *natural-language structural obfuscation* (splitting or spacing a literal target), not white-box token optimisation.

Contagion and thresholds in networks. Treating propagation as contagion over a graph is classical, beginning with the SIR formulation [22]. Threshold models make an individual’s adoption a function of how many neighbours have already adopted [15, 32]; epidemic models supply the complementary criterion that spread is subcritical when the spectral radius of the next-generation matrix falls below one, a threshold that can vanish on heavy-tailed topologies [20, 27, 28]; small-world structure [33] makes cascades tractable, and robustness to node failure follows from the error-tolerance literature [2]. Controlled experiments show that *complex* contagions—requiring several concurrent exposures—spread more slowly and reach fewer nodes [8]: the network-level reason to expect per-edge survival, not per-agent susceptibility, to govern reach. Seeding nodes to maximise reachability is NP-hard in general [21] but linear-time on acyclic graphs by topological traversal, which is why Eq. (3) is exact on our networks. This is the lens recent theory work on multiplex Markov contagion in agent swarms adopts [23]. Our results sharpen where the analogy fails for LLM agent networks: for strictly acyclic feed-forward graphs the standard linearized next-generation matrix is nilpotent ($\rho(M) = 0 < 1$) as a structural property of acyclicity rather than an empirical safety guarantee (Section 3.8); the empirical reproduction number can be below one while reachability stays high; and the per-hop rates such an analysis consumes do not transport from isolated measurement (the transportability depends on graph structure and role/context as discussed in Section 2.1).

2.1 Scope and assumptions

We clarify the scope of our theoretical claims.

Independence and Eq. (3). The product-of-complements form $1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u])$ assumes that, conditional on upstream states, the events by which different parents contribute reachability to v are independent (equivalently, parent-reach events are conditionally independent given graph structure). A sufficient condition for this identity to hold exactly is that the upstream paths from entry to different parents do not induce correlation that affects v ’s reachability (e.g. the relevant ancestors are disjoint or the parent-reach events factor appropriately). When parents share common ancestors (and thus their reachability paths overlap), these events become correlated and the product-of-complements is an approximation rather than an exact identity; in that case a path-based formulation with inclusion–exclusion (or explicit correlation terms) is required. We do not claim the identity holds for arbitrary DAGs with shared ancestors without such adjustment. Our claims about composition/transport and percolation are restricted to the feed-forward graph classes evaluated in this study (chains, star/tree-like topologies with a single entry), not to all general networks or to arbitrary DAGs. Throughout, we state assumptions explicitly rather than asserting the identity “in general”.

Reproduction number R_0 . We report the empirical reproduction number R_0 as a *descriptive, finite-horizon amplification statistic* for the finite acyclic graphs we evaluate. We operationally define R_0 as

the mean number of downstream neighbours newly compromised per compromised agent-instance, with generation convention, denominator (compromised agent-instances), time horizon (finite run length), and scope (acyclic graph class) specified. For finite feed-forward (strictly acyclic) networks, the standard next-generation mean-offspring matrix is strictly triangular and hence nilpotent with spectral radius $\rho(M) = 0$; this is a structural property of acyclicity, not an empirical safety threshold. Accordingly, we do not treat $\rho(M) < 1$ as a safety criterion here and avoid interpreting R_0 in that asymptotic threshold sense; its role is descriptive/heuristic for the evaluated finite setting.

Parallel paths and defence placement. Defence placement claims are limited to the graph classes we tested. Effectiveness depends on whether a hardened node lies on all s - t paths (i.e. on a *minimal vertex/edge cut*) rather than on “being on any path”. When there exist parallel or edge-disjoint paths to the target, hardening a node on only one path cannot block the others; we do not assume uniform effectiveness across parallel paths. Where paths are unique, effects are path-specific.

Transportability. The equality $s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}$ (transportability of isolated per-edge estimates to in-chain propagation) is treated as an empirical hypothesis to be checked for each graph class and model, not as a general theorem. Findings generalising beyond the evaluated graph classes (chains, star/tree-like feed-forward topologies with the specified structure) should be made cautiously.

Outcomes and judge. Our primary outcome is *textual propagation* of the target (presence of target string/code fragments) as measured by ASV (deterministic). Distinguishing instruction *adoption/compliance* from mere textual propagation requires additional validation; we do not conflate these levels in the main claims and note judge scope in the evaluation.

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0 , following InjecAgent’s indirect channel [38]. We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational task), so that an agent which resists injection merely by refusing to work is not credited as a defence. Injected content is therefore framed as benign-looking infrastructure text rather than as an overt command; in a pilot this framing was the difference between 0% and 60–100% compliance.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates—controlled per-hop success reaches 0.94 on one local model—so all headline numbers use Family B.

Family B (semantic instruction competition) makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the open interval $(0, 1)$ on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [26]: $C = 1[\text{ASV} \geq \tau_{\text{asv}} \vee \text{MR} \geq \tau_{\text{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\text{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment MR variant scored high similarity on clean benign outputs and on refusing outputs alike—assigning credit precisely to the outputs a defence is supposed to produce—whereas the ASV-only rule separates them cleanly (calibration accuracy 1.000 versus 0.795 on one local model; supplementary Appendix E, Table 5, states the scope of that calibration). Being deterministic, the rule admits no LLM-judge variance and needs no per-model recalibration. We clarify outcome levels: (a) *textual propagation*—the target string/code fragments appear (measured by ASV); (b) *instruction adoption/compliance*—the agent acts in accordance with the injected instruction (not merely quoting); (c) *harmful action*—the agent performs the corresponding harmful action. ASV primarily measures (a). In this paper we report (a) as the main propagation outcome; we do not automatically equate (a) with (b) or (c), and we distinguish these levels in terminology throughout. Targeted validation of a subset (to separate adoption from verbatim transmission/benign references) and the judge’s scope and limitations are detailed in the supplementary materials.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i . For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u ’s compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v . Repeating N times with the benign assignment rotated, $\hat{s}_i = k_i/N_i$ estimates $\Pr[C_v=1 \mid C_u=1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction

and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact per trial. Reusing one artefact makes $\hat{s}_i$ inherit a single sample’s idiosyncrasy, and since the product is built from those same estimates the error does not cancel: it produced a significant apparent deviation in *our own* first measurement that vanished once artefacts were drawn per trial (Section 5.3).

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl} , survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat} , survival measured *within the propagation process*, $\Pr[C_i=1 \mid C_{i-1}=1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i=1 \Rightarrow C_{i-1}=1$, and therefore

$$\text{ASR} = \Pr[C_k=1] = \prod_{i=1}^k \Pr[C_i=1 \mid C_{i-1}=1] = \prod_{i=1}^k s_i^{\text{nat}} \quad (1)$$

is an *identity*, not a modelling assumption, and it holds empirically: for one frontier model, $\prod_i s_i^{\text{nat}} = 0.850 = \text{ASR}$. The claim usually described as “the Markov product assumption” is, stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}. \quad (2)$$

This is the assumption single-agent injection benchmarks make implicitly, and it is what we test.

3.6 Reproduction number and utility

We report the empirical reproduction number R_0 (mean new compromises per compromised agent-instance) with a Wilson interval on the underlying compromise proportions, alongside its regular-topology target $d \cdot \bar{s}$ (d = mean out-degree) as a consistency check, not an identity. Because a network can achieve $\text{ASR} = 0$ by refusing to process any untrusted content, we also measure the paired legitimate-task performance $\Delta U = U_{\text{clean}} - U_{\text{attack}}$ and Retention = $U_{\text{attack}}/U_{\text{clean}}$, running each configuration twice (with and without injection). Here U indicates that the final legitimate answer is uncontaminated by the attack target; we scope the metric accordingly (Section 7).

3.7 Estimators

All proportions use Wilson score intervals [34], which we prefer to the Wald interval because many of our estimates sit at or near 0 and 1 [6]. For the composition test we use a bootstrap [14] test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$: because ASR and the $\hat{s}_i$ come from independent experiments, variances add and an independent bootstrap over both sources is valid. We report the point estimate, a percentile interval, a two-sided p -value for $H_0 : \Delta = 0$, and the minimum detectable effect $\text{MDE} = (z_{1-\alpha/2} + z_{\text{power}}) \text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*. Multiplicity across cells

is controlled by Benjamini–Hochberg [5] at $q = 0.05$; Section 5.9 reports which cells survive it, and why we decline to read any small- n null as evidence of transport.

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e , the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u]), \quad \Pr[v_0] = 1. \quad (3)$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.7). The assumption in Eq. (3) is conditional independence of *edge activations* given the upstream state and the graph, so the naive product $\prod (1 - p_i)$ holds when the parents’ upstream paths are disjoint and no unmodelled shared ancestor induces correlation between the parent reachability events; it does *not* hold for arbitrary DAGs in which parents share ancestors, where a path-based (inclusion–exclusion) formulation or an explicit correlation term is required. We therefore treat the product-of-complements form as exact under the stated graph assumptions and as a marginal percolation approximation otherwise, for the acyclic classes we evaluate (chains and the specified tree structures); per-context heterogeneity of a single edge does not by itself invalidate a marginal formulation, and variance and clustering are handled in estimation (Section 2.1).

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M , with $M_{uv} = s_{u \rightarrow v}$, satisfies $\rho(M) < 1$ [28]. For a feed-forward network this criterion is *vacuous*. Ordering nodes topologically makes M strictly triangular, so every eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with $\text{ASR} = 0.450$, and $R_0 = 0.420$ on the seven-agent star with $\text{ASR} = 0.725$: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0 . Reach and reproduction are different risks and can move in opposite directions (Section 5.6). We therefore define R_0 operationally—the mean number of downstream neighbours newly compromised per compromised agent-instance over the observed finite horizon, with generation convention, denominator and unit of analysis stated—and use it as a *descriptive, finite-horizon amplification statistic*: because $\rho(M) = 0$ here follows from acyclicity rather than from measurement, we do not read it as a safety threshold, and the quantity a practitioner cares about is reachability, Eq. (3).

A corollary concerns defence placement, which in feed-forward graphs depends on paths and cut sets rather than on any uniform property of nodes. Take the s – t DAG with edge-disjoint paths $s \rightarrow A \rightarrow t$ and $s \rightarrow B \rightarrow t$: hardening A blocks only the first, so reachability can remain non-zero and only a cut intersecting every s – t path (here $\{A, B\}$) blocks the target. Effectiveness is thus conditional on lying on a minimal s – t cut, and where a unique path exists it is path-specific; we therefore avoid general “equal effectiveness” claims and treat placement as case-specific for the tested graph classes. On our measured rates this is visible directly: on the single-path chain every interior node reduces predicted ASR by 0.450 (to zero), while on the tree hardening agent_2 (on the entry–target path) takes ASR from 1.000 to 0.000 but hardening agent_1 (another branch) changes nothing.

4 Experimental Setup

Models. Five models from five independent developers: qwen2.5:7b (Alibaba, local), DeepSeek V3.2 (DeepSeek), Llama 3.3 70B (Meta), Nova Pro (Amazon), and Claude Sonnet 4.5 (Anthropic); the latter four via Bedrock, region us-east-1. Temperature 0.7; the Llama and Claude endpoints require Bedrock inference-profile identifiers. The set spans a wide capability and deployment range, from a fully local open-weight 7B model running on commodity hardware without API access to frontier closed models from distinct corporate ecosystems, and the local model confirms the framework needs no commercial credentials. Using five *independent* developers also minimises the risk that a shared fine-tuning corpus or RLHF pipeline explains cross-model patterns.

Agent roles and network wiring. Each node is assigned a role—*planner*, *worker*, *reviewer*, or *aggregator*—and receives a fixed system prompt encoding that role’s responsibilities: the planner (agent_0, always the entry node) coordinates, the worker (agent_1) executes a delegated sub-task, the reviewer (agent_2) inspects and summarises incoming content, and the aggregator (agent_3) synthesises a final answer. These roles reflect the structural positions that appear in real orchestration frameworks [13, 18, 35] without over-specifying the prompt engineering. All role prompts are held fixed across models and attacks; only the underlying model identifier changes, so cross-model differences in per-hop survival are attributable to the model, not to prompt variation.

Cells. Unless stated otherwise: four-agent chain, entry agent_0, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$. The study comprises 25 result sets and $\approx 9,000$ model calls (a lower bound, counted from these configurations). Benign assignments are drawn from a pool of three short operational questions (summarise, fact-recall, draft) rotated cyclically, so no trial sees the same context twice and the three contexts are semantically orthogonal to the injected task (Section 5.9 shows why this matters).

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (BANANA – 77, requiring recombination), or

Table 1: Chain, no defence, Task Family B; 40 natural runs and 30 controlled trials per edge. $\bar{s}$ is the mean controlled per-edge survival; per-edge order follows the chain. Rows marked $\dagger$ used the fresh-artefact protocol (Section 5.3).

Model	ASR	$\bar{s}$	per-edge
Llama 3.3 70B $\dagger$	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 $\dagger$	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

split (BANANA and 77, combine with a hyphen). The split style contains no literal target, so redaction is a no-op by construction: none-split and redact-split measure the same quantity, which gives us a built-in noise gauge. Defence scope is deliberately narrow—a static, non-adaptive attacker and two rule-based defences; adaptive attackers and learned detection layers [9, 37] are left for future work.

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth: Wilson interval coverage over a grid of p and n ; size and power of the composition bootstrap test against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree. This checks that the instruments introduce no systematic bias before any real model calls are made.

5 Results

5.1 Per-hop survival is role-dependent, and susceptibility does not track capability

Table 1 reports the per-model chain headline. Two facts stand out. First, per-hop survival varies several-fold *across receiver roles within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867, so a scalar s per model is unjustified. Second, susceptibility does not follow capability: the two frontier commercial models sit at the bottom of the panel and a 70B open-weight model at the top, so ASR does not follow model scale, and the five models yield four qualitatively different outcomes. Both facts point the same way: a single end-to-end number per model discards *where* in the network the attack was stopped.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on the two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75 $\times$ underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$ against ASR = 0.850, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$.

For *DeepSeek V3.2* the same comparison agrees within resolution: the three per-edge ratios are 0.90, 0.99 and 1.10, $\prod_i s_i^{\text{ctrl}} = 0.409$

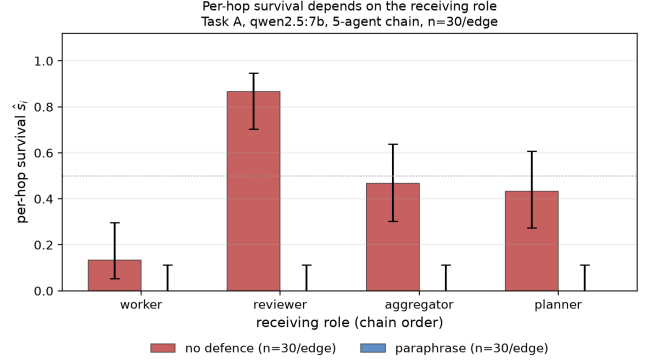


Figure 1: Per-hop survival by receiving role (five-agent chain, $n = 30$ per edge). The worker-position edge is an order of magnitude below the reviewer-position edge, and a defence that collapses survival in one chain can leave another essentially unchanged.

Table 2: Transportability of the isolated per-hop estimate, per edge and per model; 40 natural runs and 30 controlled trials per edge. *ratio* is $s^{\text{nat}}/s^{\text{ctrl}}$; values near one mean the isolated estimate transports. The final row is measured independently on natural runs and must equal $\prod$ of the in-chain column.

model	edge	s^{ctrl}	s^{nat}	ratio
Llama 3.3 70B	a0→a1	1.000	1.000	1.00×
	a1→a2	0.233	0.875	3.75×
	a2→a3	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	—
	$\prod$ (in-chain) / ASR	—	0.850	—
DeepSeek V3.2	a0→a1	0.667	0.600	0.90×
	a1→a2	0.800	0.792	0.99×
	a2→a3	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	—
	$\prod$ (in-chain) / ASR	—	0.400	—

against ASR = 0.400, and $\Delta = -0.009$ $[-0.225, +0.201]$, $p = 0.930$ —a *non-rejection* at MDE ≈ 0.26 ($n = 40$), not distinguishable from exact. In both cases Eq. (1) holds exactly, as it must.

The practical reading is that isolated per-hop survival—the regime single-agent injection benchmarks inhabit—can misestimate in-chain survival by a factor of four, but does not always; whether it does is an empirical property of the model and edge.

5.3 Protocol sensitivity: a significant result that did not replicate

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i s_i^{\text{ctrl}} = 0.806$ against ASR = 0.475: a large apparent deviation in the opposite direction, with $\Delta = -0.331$, 95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact versus a fresh artefact per trial (Llama 3.3 70B and DeepSeek V3.2, chain, 40 natural runs, 30 controlled trials per edge). The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	policy	per-edge	Π	Δ	p
DeepSeek	fixed	0.967/1.000/0.833	0.806	−0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	−0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

Table 4: Content-form ablation (Llama 3.3 70B, chain of $n = 4$ agents; 40 natural runs, 30 replays per arm and edge). Both replay arms use the same receivers, assignments and judge, so the difference is attributable to payload form alone.

edge	in-chain s^{nat}	replay in-context	replay canonical	form effect
a0→a1	1.000	1.000	1.000	0.000
a1→a2	0.950	0.967	0.067	+0.900
a2→a3	0.947	0.733	1.000	−0.267

with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \rightarrow +0.617$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact. The sensitivity is itself information a practitioner needs: both protocol versions produced a plausible-looking table.

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion—whereas in natural runs the same target travels inside the sender’s legitimate work product and may be consumed as content rather than as a command. To separate *form* from *regime* we replay both content sources through the *same* harness: a fresh receiver, the same rotating assignment, the same judge. The harness is self-validating, because one arm replays content captured from real natural runs and must reproduce s^{nat} .

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 embedded in a work product and at 0.067 as a bare assertion—a 14× difference produced entirely by packaging. We read this as the mechanism behind Section 5.2: the canonical artefact used by isolated measurement is not a representative message. The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733), so form interacts with position, and there the in-context replay does not reproduce the in-chain rate (0.733 versus 0.947)—the harness self-check passes on two of three edges, so we do not treat the third as a clean comparison.

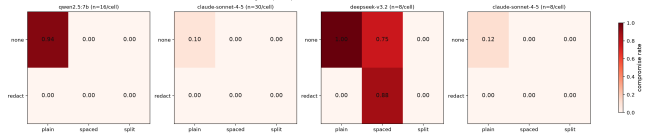


Figure 2: Single-hop compromise rate by defence and attack form, one panel per model ($n = 30$ per cell for Claude 4.5, $n = 8$ for DeepSeek, $n = 16$ for qwen; we do not interpret small differences in the $n = 8$ cells). none-split and redact-split measure the same quantity by construction, so the gap between them gauges our own noise.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Fig. 2) shows why this is weaker than it appears. On the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88: it works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack succeeds only 0.10 of the time even with no defence, so redaction’s apparent perfection is the absence of anything to defend (invariant for $\tau_{\text{ASV}} \in \{0.7, 0.8, 0.9, 1.0\}$, supplementary Appendix C). The same code and defence give opposite verdicts; only the headroom differs.

Semantic paraphrase is worse than useless in our cells: on the model where the attack succeeds it leaves ASR unchanged ($0.450 \rightarrow 0.450$) while reducing utility retention from 0.600 to 0.400, so it is *strictly dominated* by doing nothing; on the resistant model it still costs 0.100 of utility where ASR is already 0.000. This matches prior single-agent findings that paraphrasing is breakable [37].

Both verdicts cover two defences against a static, non-adaptive attacker, and the small panels serve only the within-model contrast.

5.6 Topology changes the spread of survival, not just the amount

Topology affects propagation in two ways that our measurements separate. The first is structural and expected: out-degree and entry-point position change the reproduction number. On a seven-agent star with the entry at a leaf, only one edge carries the attack; measured on Llama 3.3 70B the centre is compromised with $\text{ASR} = 0.725$, giving $R_0 = 0.420$ against a regular-topology target of $d \cdot \bar{s} = 0.652$. The gap is structural, not noise—the aggregator has no successors—and it reproduces the deviation our synthetic calibration predicted for star fan-in.

The second effect is less obvious and does not follow from the first. Table 5 compares three seven-agent topologies with nearly the same mean per-hop survival yet sharply different outcomes. The star reaches its aggregation point more often than the chain ($\text{ASR} = 0.725$ versus 0.450) while reproducing less ($R_0 = 0.420$ versus 0.821): reach and reproduction are not the same risk, and a topology can move them apart. The tree is most permissive on both measures ($\text{ASR} = 0.800$, $R_0 = 0.831$) and the only cell in which

Table 5: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge). The three topologies have similar mean per-hop survival but differ substantially in outcome.

topology	edges	$\bar{s}$	ASR	R_0
chain	6	0.761	0.450	0.821
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	0.800	0.831

R_0 approaches its regular-topology target ($d \cdot \bar{s} = 0.795$, within 5%), the approximation holding where branching is genuinely balanced.

A third effect concerns dispersion: chain edges connect several receiver roles and survival spans 0.27 to 1.00, whereas all six star edges connect a worker to the aggregator and survival is nearly homogeneous (0.667–0.867). A topology therefore does not merely multiply a common per-hop rate by a degree-dependent factor—it determines how much role heterogeneity the network averages over, and hence whether a single scalar s summarises it at all. Repeated on Nova, DeepSeek and a second Llama instance ($n = 10$), the star/tree pattern survives wherever propagation is non-zero (DeepSeek 0.75 versus 0.475; second Llama 0.675 versus 0.65) with Nova at the resistance floor.

5.7 The shape of the depth profile is a diagnostic

The relative error of the isolated product grows with depth at a rate that is model-specific, and that rate—not the error at any single depth—is the usable diagnostic. Across models the slopes form a *spectrum* rather than two categories: +11.8 points per hop on qwen2.5:7b ($\rho = +1.00$), +6.8 on Llama 3.3 70B ($\rho = +0.77$, reproduced at higher power) before its chain dies at depth 11, and +2.3 on DeepSeek V3.2 at $n = 10$ ($\rho = +0.65$)—its apparent flatness at $n = 8$ ($\rho = -0.07$) is a reminder that a short low-power chain can read as flat. The ordering follows from what Section 5.2 measures directly: where the isolated rate transports best (DeepSeek matches its measured ASR to 2.2%) composing edge estimates adds little systematic error, whereas an edge under-estimated by 3.75× (Llama’s middle edge) enters every product containing it and compounds with depth. The profile’s *slope* therefore diagnoses transportability while its *direction* stays model- and topology-dependent—on the tree the isolated estimate is optimistic, predicting certain compromise against a measured 0.800. Independence is not what fails: evaluated on in-chain rates, Eq. (3) reproduces the measured ASR in all three topologies. The per-model table and rank statistics are in the supplementary material.

5.8 Recurrence makes the threshold non-vacuous

Section 3.8 shows the threshold criterion vacuous for acyclic graphs and informative only once the interaction graph contains a cycle. We test that second half on a manager-with-reporting-workers pattern: the spectral radius reduces to $\rho(M) = \sqrt{\sum_j a_j c_j}$, yielding *opposite* predictions borne out by measurement—the Llama and DeepSeek loops are supercritical ($\sum_j a_j c_j = 1.800$ and 1.018) while qwen’s decays (0.578; supplementary Appendix D).

5.9 Utility and estimator validation

Reporting a defence’s propagation rate without its cost is uninformative, and our estimator claims rest on synthetic checks that belong in the appendix. The $(ASR, \Delta U)$ pairs for three defences on two models—redaction blocks propagation at *no* utility cost on the susceptible model while paraphrase buys no benefit and costs utility—together with Wilson coverage, the size and power of our bootstrap test, and the sample sizes implied by the MDE, are in the supplementary material; in short, $n = 40$ excludes only deviations above 0.26, so we treat no $n = 40$ null as evidence and label the degenerate $ASR = \prod_i \hat{s}_i = 0$ cell uninformative.

6 Practical Implications

The four prescriptions of Section 1 change what is *reported* rather than what is *built*—composition is safe once transportability is checked, defence results need a matched baseline to have headroom, and placement follows entry–target paths and cuts—and each is derived in full in the supplementary material.

7 Limitations and Threats to Validity

Our coverage is deliberately narrow—one benign payload, a static non-adaptive attacker, four to fifteen agents, cell sizes of 30–40 (MDE ≈ 0.26) and two defence families—the utility metric is a contamination proxy rather than a correctness oracle, and the transportability failure we can power is real but its prevalence unknown. Threats to validity, including the protocol sensitivity that flipped one of our own results ($p = 0.0030 \rightarrow 0.930$ under fresh artefacts) and the scope of the judge, are itemised with their measurements in supplementary Appendices E and H.

8 Conclusion

A compromise’s reach in an LLM agent network is a property of the graph, the roles travelled and the form in which the payload is re-packaged at each hand-off, not of any single agent. Separating the identity $ASR = \prod_i s_i^{\text{nat}}$ from the transportability hypothesis $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$ gives four results: composition holds whenever the rates are measured *in the chain*, and the failure mode—transporting isolated estimates, detected by the depth-error slope—is what single-agent benchmarks implicitly assume away; payload form is a first-order mechanism (0.067 bare versus 0.967 embedded); the epidemic threshold is identically satisfied on any acyclic graph and cannot certify safety, so reachability comes from Eq. (3) rather than R_0 ; and defence efficacy is unidentifiable without a matched baseline, with placement set by graph structure rather than by measured per-hop survival. Adopting these prescriptions changes what is reported rather than what is built, at the cost of one long-chain probe.

Ethical Considerations

All experiments use a *benign* marker payload against our own harness via standard provider APIs; no real user data or third-party systems were involved.

Artifact Availability

Code, per-cell outputs and offline re-analysis reproduce every table with no model call; a manifest maps each table to its command.

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